

WATER PUMP CONTROLLER

■ SURESH KUMAR K.B.

A water pump controller senses the level of water in a tank and drives the water pump.

The circuit described here is built around timer IC1 (555).

When the water level of tank goes below the low level marked by 'L' the voltage at pin 2 of IC1 becomes low. As

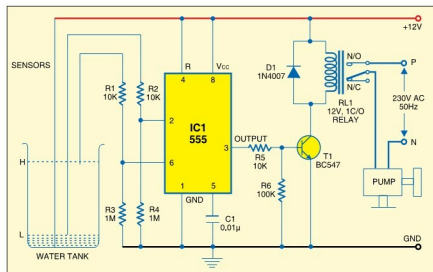
a result, internal SR-flip-flop of IC1 resets and its output goes high. This high output pin 3 of IC1 drives relay driver transistor T1 (BC547) and energises relay RL1. Water pump gets mains power supply through N/O contacts of the relay and is powered on. It starts filling water in the tank.

When the water level goes high and reaches the high level marked



by 'H', the voltage at pin 6 of IC1 also goes high. As a result, internal SR-flip-flop of IC1 sets and its output goes low. This low output pin 3 of IC1 cuts off relay driver transistor T1 and de-energises relay RL1. Water pump disconnects from the mains power supply through N/C contacts of relay and goes off. This stops the water flow.

Use 12V battery or 12V adaptor to operate the circuit. Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Use three-sensor (thick conductive wire) and fit into the tank ensuring that 12V sensor touches the bottom and low-level sensor remains just above the bottom. High level sensor should be placed at a height till where the water needs to be filled. The circuit is economical, highly reliable and can be easily constructed. ●



ELECTRONIC CANDLES

■ RAJ K. GORKHALI

Here is a simple circuit that can produce the effect of candle light in a normal electric bulb. A candle light, as we all know, resembles a randomly flickering light. So, the objective of this project activity is to produce a randomly flickering light effect in an electric bulb.

To achieve this, the entire circuit can be divided into three parts. The first part comprises IC1 (555), IC2 (74LS164), IC3 (74LS86), IC4 (74LS00) and the associated components. These generate a randomly changing train

of pulses.

The second part of the circuit consists of SCR1 (C106), an electric bulb connected between anode of SCR1 and mains live wire, and gate trigger circuit components. It is basically half-wave AC power being supplied to the electric bulb.

The third part is the power supply circuit to generate regulated 5V DC from 230V AC for random signal generator. It comprises a stepdown transformer (X1), full-wave rectifier (diodes D3 and D4), filter capacitor (C9), followed by a regulator (IC5).

The random signal generator of the circuit is built around an 8-bit

serial in/parallel out shift register (IC2). Different outputs of the shift register IC pass through a set of logic gates (N1 through N5) and final output appearing at pin 6 of gate N5 is fed back to the inputs of pins 1 and 2 of IC2. The clock signal appears at pin 8 of IC2, which is clocked by an astable multivibrator configured around timer (IC1). The clock frequency can be set using preset VR1 and VR2. It can be set around 100 Hz to provide better flickering effect in the bulb.

